

WindFormer: A Common-Speed and Direction-Vector Network for Wind Resource Forecasting

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Abstract

Accurate short-term wind resource forecasting is critical for wind farm dispatch and power system operation, yet wind speed and wind direction exhibit different cross-turbine structures and mathematical properties. Wind speed combines shared meteorological evolution with turbine-level deviations, whereas wind direction is turbine-specific and circular, making a single homogeneous interaction pattern inadequate. We propose WindFormer, a structure-aware dual-branch network that extracts a learnable common wind-speed field, models turbine residuals with shared predictors, and organizes the common history using Phase-Attention over screened candidate temporal scales. In parallel, WindFormer forecasts each turbine’s wind direction in a continuous sine-cosine vector space with shared Patch-Attention, and couples the two branches through a physical wind-vector consistency objective. Experiments on two public wind-farm datasets show consistent improvements over the compared forecasting baselines across wind-speed and wind-direction

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metrics and forecast horizons. Component ablations, matched-seed statistical tests, perturbation and extreme-volatility stress tests, efficiency measurements, and sparse-deployment zero-shot experiments further examine the mechanism, stability, computational cost, and cross-turbine generalization of the proposed framework.

Keywords: Spatio-temporal modeling, wind speed forecasting, wind direction forecasting, wind turbines, deep learning, time series forecasting

1. Introduction

Driven by its clean nature and reliable supply, wind power has become an important energy source for modern power systems. Accurate short-term forecasts of wind speed and wind direction support grid dispatch, resource allocation, and wind-energy utilization [1–4]. In a wind farm, however, these variables are not homogeneous multivariate channels. They are shaped by different physical structures across turbines and also have different mathematical properties [5, 6]. A forecasting model should therefore share information selectively rather than imposing one interaction pattern on every wind variable.

1.1. Challenge 1: Shared but Non-identical Wind-Speed Dynamics

Wind turbines in the same farm are driven by common weather processes and large-scale incoming flows, so their wind-speed trajectories often exhibit synchronized trends. At the same time, local terrain, wake effects, and turbine positions produce turbine-level deviations [7, 8]. This combination creates a structural tension: channel-independent methods preserve

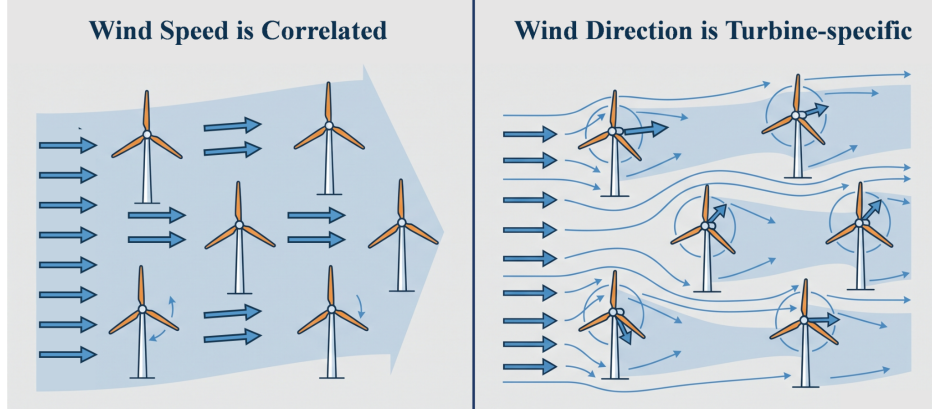


Figure 1: Conceptual contrast between shared but non-identical wind-speed dynamics and turbine-specific wind-direction variation.

local freedom but repeatedly relearn common variation, whereas fully correlated methods may mix common dynamics and local deviations in a single representation. The key challenge is therefore to share the common wind field while retaining turbine-specific residual information.

Figure 2 provides supporting evidence for this view. Representative turbines exhibit similar dominant spectral components and wavelet-energy patterns, suggesting a shared temporal scaffold rather than a collection of unrelated sequences [9–11]. The remaining differences motivate a separate residual branch instead of assuming that all turbines are identical [12].

After separating the shared component, the common wind-speed history still needs an effective temporal organization. Rather than treating a screened candidate scale as a strict physical period, we organize the history by candidate temporal scales and relative phase positions within temporal blocks [10, 13–15]. This phase-aware organization allows local dynamics at comparable positions across blocks to interact.

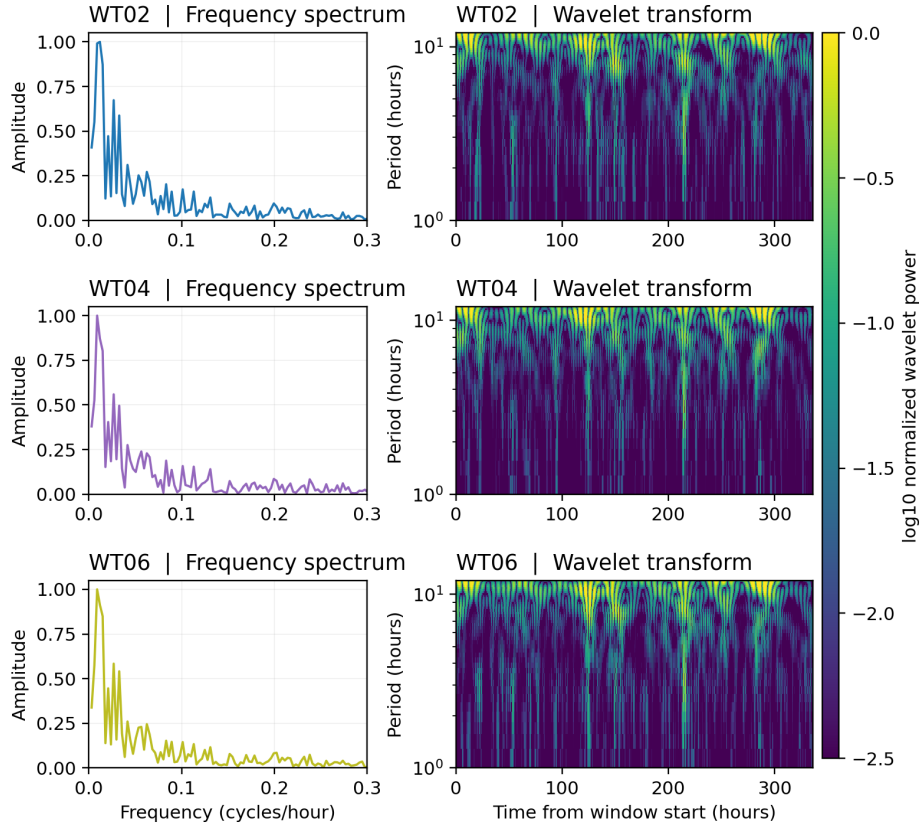


Figure 2: Frequency-spectrum and wavelet-transform comparison of representative turbine wind-speed sequences. Similar dominant components support a shared temporal scaffold, while turbine-level deviations motivate residual modeling.

1.2. Challenge 2: Turbine-specific and Circular Wind Directions

Wind direction follows a different cross-turbine structure. Interactions among turbine locations, wake effects, and local flow fields cause different units to exhibit distinct dominant directions and turning patterns, so direction should not be collapsed into a common representation in the same way as wind speed. Figure 3 (a) shows that the dominant directions and directional concentration levels of different turbines are not consistent. Figure 3 (b) reports that about 51% of turbine pairs have an average angular distance of no less than 90° , indicating that the heterogeneity is not caused by a few isolated units. Figure 3 (c) further displays structured variation in the pairwise angular-distance matrix. These observations motivate turbine-wise direction modeling.

Wind direction is also a circular variable rather than an ordinary one-dimensional scalar [16–18]. It satisfies

$$\theta \equiv \theta + 360^\circ. \quad (1)$$

Consequently, 359° and 1° are physically close even though their scalar difference is 358° . Direct scalar regression can therefore create artificial discontinuities in distances, local variations, and error gradients. To preserve this geometry from input to prediction, WindFormer maps each direction angle into a two-dimensional sine-cosine unit vector:

$$\mathbf{u}_{t,n} = [\cos(\theta_{t,n}), \sin(\theta_{t,n})]. \quad (2)$$

The model predicts in this continuous representation, applies direction-vector and wind-vector consistency constraints, and recovers the angle using

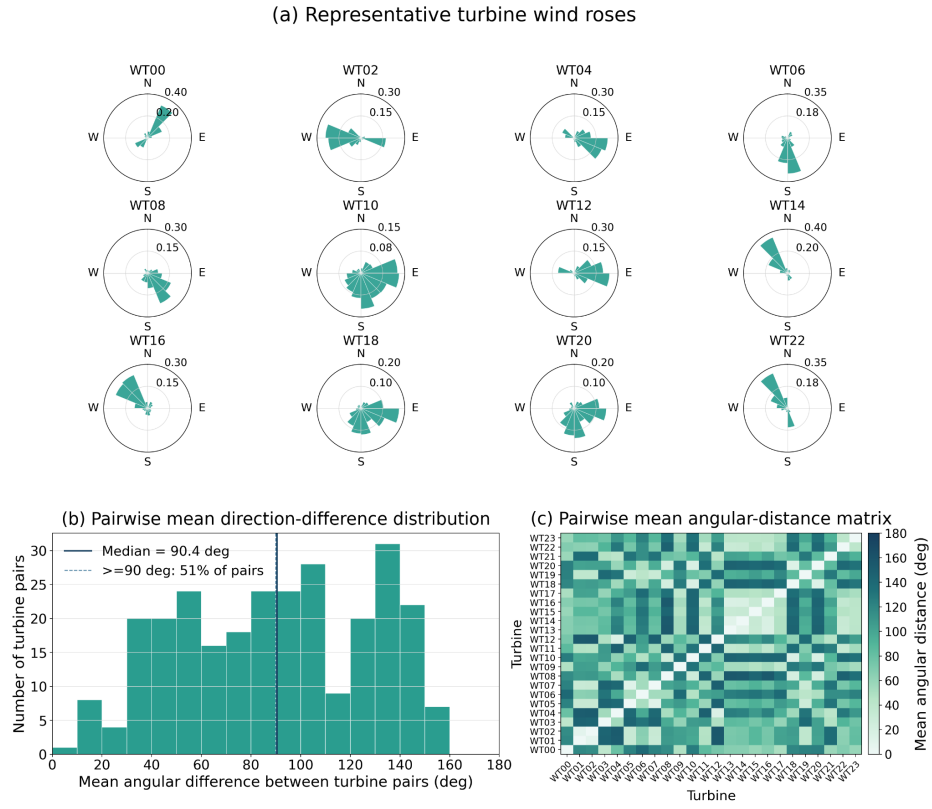


Figure 3: Turbine-specific wind-direction distributions and all-pair angular differences among 24 turbines.

$$\hat{\theta}_{t,n} = \text{atan2}(\hat{u}_{t,n}^{\sin}, \hat{u}_{t,n}^{\cos}). \quad (3)$$

Because directional patterns remain turbine-specific, each turbine is processed as an individual two-dimensional sequence. The temporal patch projection and Patch-Attention encoder are shared across turbines, allowing transferable local dynamics to be learned without collapsing distinct directional distributions into one global state.

1.3. *WindFormer Framework*

The two challenges lead to a single design principle: share information where the wind field is physically common, preserve information where turbine behavior is specific, and keep the direction representation geometrically valid. WindFormer implements this principle with two representation-decoupled branches and one objective-level coupling mechanism. The wind-speed branch extracts a learnable common field, models it with Phase-Attention organized by screened candidate temporal scales, and reconstructs turbine forecasts from the common prediction and shared residual predictors. The wind-direction branch retains each turbine’s sequence, maps angles to sine-cosine components, and uses shared Patch-Attention to forecast future direction vectors. The branches are trained jointly through wind-speed, direction-vector, and wind-vector consistency losses.

The main contributions of this paper are as follows:

- **We separate shared and local wind-speed dynamics.** A learnable aggregation extracts shared wind-speed dynamics, while a shared

residual branch restores turbine-level deviations. A phase-aware temporal encoder organizes the common history by screened candidate scales and relative positions rather than assuming a strict physical period.

- **We develop a turbine-wise circular direction branch within the same forecasting framework.** Sine-cosine vectors preserve boundary continuity, and shared Patch-Attention captures transferable temporal patterns while retaining turbine-specific directional sequences.
- **We couple the two task-specific branches through a physical wind-vector objective and evaluate the resulting WindFormer framework comprehensively.** Standard forecasting, component ablations, robustness tests, sparse-deployment transfer, efficiency measurements, and matched-seed statistical tests together assess accuracy, mechanism, stability, and generalization.

2. Related Work

2.1. Spatial Relationships in Multi-Turbine Wind Power Forecasting

Existing deep learning studies usually improve wind forecasting by modeling spatial relationships among turbines. Graph-based and convolutional recurrent models encode turbine topology, neighborhood influence, multi-scale spatial dependence, or local and long-range interactions into the forecasting process [19–22]. To handle condition-dependent relationships, later studies further introduce channel-aware dependence and temporal lead-lag

structures [23–26]. These methods show that cross-turbine information is useful for short-term wind forecasting.

Generic multivariate time-series models also provide useful designs for organizing turbine variables. PatchTST reduces cross-channel interference through channel-independent temporal modeling [27], TimesNet converts temporal variations into two-dimensional period structures [28], iTransformer learns dependencies along the variable dimension [29], and MSGNet enhances multi-scale cross-sequence interaction [30][13, 14, 31, 32]. MST-Net further discusses the combination of channel-independent and spatially correlated modeling for short-term wind resource forecasting [33, 34]. More recently, an Information Sciences study further explores a global-local dual-branch design and pyramid patch interaction for multivariate forecasting [35].

However, most existing methods regard cross-turbine relationships as a single interaction pattern to be learned. They seldom separate the common wind-speed field shared by multiple turbines from turbine-specific residual variations. This distinction is important because wind speeds in the same wind farm often share dominant meteorological trends, while local terrain, wake effects, and turbine positions still introduce individual deviations.

2.2. Wind Direction Forecasting and Two-Dimensional Representation

Wind direction forecasting has been studied together with wind speed prediction, yaw-state modeling, and short-term directional change prediction. Representative works include ARMA-based joint wind speed and direction forecasting [36], yaw-error prediction based on wind-direction changes [37], and two-phase deep models designed for short-term wind-direction sequences [38].

Unlike wind speed, raw wind direction is a one-dimensional angle with boundary discontinuity. Sine-cosine representations provide a natural way to embed direction angles into a continuous two-dimensional space [36, 37]. Nevertheless, existing studies mainly focus on single direction sequences or general speed-direction joint prediction. Systematic modeling of multi-turbine directional heterogeneity under periodic-boundary constraints remains insufficiently explored.

3. Data Preprocessing and Task Definition

3.1. Dataset Sources and Wind Farm Scenarios

This paper studies short-term joint forecasting of wind speed and wind direction for multiple turbines. The wind resource observations are provided by supervisory control and data acquisition (SCADA) systems [39, 40]. Following common wind-farm forecasting practice, this paper uses 10-minute averaged data to preserve short-term wind variability while controlling data volume.

Two public wind farm datasets are used to evaluate the model under different layouts and terrain conditions. The first dataset was collected from the Penmanshiel wind farm in the UK. It contains 10-minute SCADA records from 14 wind turbines, with raw observations covering July 27, 2016 to June 30, 2021. The Penmanshiel turbines are distributed over highland and ridge areas, where common incoming flows favor shared wind-speed trends and terrain effects introduce local direction differences.

The second dataset was collected from a wind farm in Shanxi Province, China. It contains 10-minute wind speed and wind direction observations

from 24 turbines, covering November 1, 2023 to May 31, 2024. The Shanxi turbines are located in undulating valley and ridge terrain, where valley winds introduce more pronounced diurnal variations. Together, the two datasets test whether the model captures shared wind-farm dynamics while preserving turbine-level differences.

3.2. Data Preprocessing

Consider a wind farm with N turbines and a sampled sequence of length T . At time t , the wind speed and wind direction of turbine i are denoted by $v_{t,i}$ and $\theta_{t,i}$. Given the past L time steps, the model predicts the states of all turbines for the next H time steps [41, 42]:

$$\mathbf{X}_{t-L+1:t} \longrightarrow \hat{\mathbf{Y}}_{t+1:t+H}. \quad (4)$$

Given a historical window length L and a forecasting horizon H , the input at each time step contains N wind-speed channels and two trigonometric direction channels for each turbine:

$$\begin{aligned} \mathbf{x}_t = & [v_{t,1}, \dots, v_{t,N}, \\ & \sin \theta_{t,1}, \dots, \sin \theta_{t,N}, \\ & \cos \theta_{t,1}, \dots, \cos \theta_{t,N}]. \end{aligned} \quad (5)$$

Thus, each time step has dimension $3N$. The model outputs future wind speed and wind direction for all turbines. Samples are chronologically split into training, validation, and testing sets with a ratio of 7 : 1 : 2.

Raw SCADA records are aligned to a common 10-minute time axis. Wind-speed values outside $[0, 25]$ m/s and wind-direction values outside

$[0, 2\pi)$ are marked as missing. Missing speeds are linearly interpolated, while directions are interpolated after conversion to sine and cosine components [43]. For wind speed, low-amplitude frequency components are removed through FFT-based denoising before sliding-window generation; incomplete windows are discarded.

Wind speed is standardized for each turbine using only the training-set statistics:

$$\tilde{v}_{t,i} = \frac{v_{t,i} - \mu_i^v}{\sigma_i^v + \epsilon}, \quad (6)$$

where μ_i^v and σ_i^v are the training-set mean and standard deviation of turbine i , and ϵ prevents numerical instability. Validation and test data use the same statistics. Predictions are transformed back to the original physical units before metric calculation.

Common-speed extraction and residual modeling are performed in the standardized space, and final evaluation is conducted in the original wind-speed units.

Wind direction is represented through two-dimensional sine-cosine components to preserve boundary continuity:

$$\mathbf{u}_{t,i} = [\cos(\theta_{t,i}), \sin(\theta_{t,i})]. \quad (7)$$

The model predicts the two direction components, applies L2 normalization, and recovers the angle by:

$$\hat{\theta}_{t,i} = \text{atan2}(\hat{u}_{t,i}^{\sin}, \hat{u}_{t,i}^{\cos}) \quad (8)$$

This representation keeps boundary-adjacent directions close in the model space and supports vector-based error calculation.

4. PROPOSED METHODOLOGY

4.1. Overall WindFormer Structure

WindFormer is designed around two physical properties of wind farms: wind speeds tend to share common meteorological evolution across turbines, whereas wind directions are recorded as one-dimensional angles and exhibit stronger turbine-specific patterns. As shown in Fig. 4, WindFormer contains a wind-speed branch and a wind-direction branch. The branches are representation-decoupled so that each variable is modeled in its appropriate space, but they are jointly optimized through a physical wind-vector consistency objective.

Given an input window, the wind-speed branch decomposes the multi-turbine speed field into a common speed component and turbine-level residual components. The common component is encoded by Phase-Attention to capture phase-aligned evolution across temporal blocks, while the residual components are predicted by a shared residual MLP to recover local deviations. The predicted common speed and turbine residuals are then reconstructed into future turbine-wise wind speeds.

The wind-direction branch maps one-dimensional direction angles into two-dimensional sine-cosine unit vectors. For each turbine, the historical two-dimensional direction sequence is partitioned into temporal patches, embedded as patch tokens, and processed by a shared Patch-Attention encoder. A direction prediction head outputs future direction components, which are

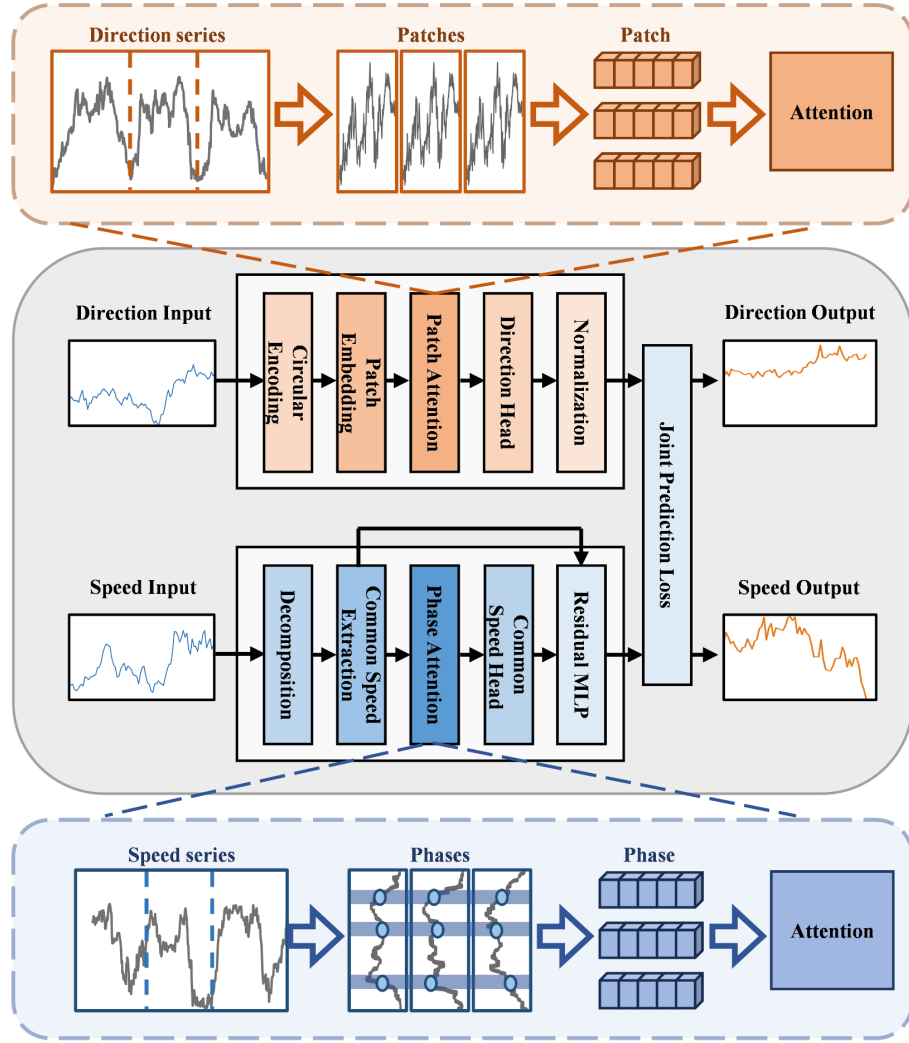


Figure 4: Overall architecture of WindFormer. The wind speed branch extracts and forecasts common wind speed dynamics with Phase-Attention and restores turbine-level residuals, while the wind direction branch preserves turbine-wise two-dimensional direction sequences through shared Patch-Attention encoding.

normalized to unit length before being converted back to angles. This branch preserves turbine-wise directional evolution while sharing temporal parameters across turbines.

The two branches output wind speed and wind direction for the same forecasting horizon. A joint objective combines wind-speed error, direction-vector error, and wind-vector consistency, thereby coupling the two physical quantities during training without forcing them into the same representation space.

4.2. Learnable Common Wind Speed Extraction

Let the standardized multi-turbine wind speed be:

$$\tilde{\mathbf{v}}_t = [\tilde{v}_{t,1}, \dots, \tilde{v}_{t,N}]. \quad (9)$$

The model sets a learnable scalar a_i for each turbine and obtains normalized weights through softmax:

$$\boldsymbol{\alpha} = \text{softmax}(\mathbf{a}), \quad \sum_{i=1}^N \alpha_i = 1. \quad (10)$$

The common wind speed sequence is defined as:

$$c_t = \sum_{i=1}^N \alpha_i \tilde{v}_{t,i}. \quad (11)$$

The corresponding wind speed residual for the i -th turbine is:

$$r_{t,i} = \tilde{v}_{t,i} - c_t. \quad (12)$$

This operation compresses shared wind-farm variations into a common sequence, while turbine-specific deviations are handled by the residual branch.

4.3. Common Wind Speed Phase-Attention over Candidate Temporal Scales

Candidate temporal-scale folding.

Let the common wind speed input sequence be:

$$\mathbf{c}_{1:L} = [c_1, c_2, \dots, c_L]. \quad (13)$$

For a screened candidate temporal scale p , the model organizes the sequence into multiple local temporal blocks according to relative phase position. The scale is a modeling choice for organizing recurring local dynamics, rather than a claim that the wind farm follows one strict physical period [13, 14]. Let:

$$K_p = \left\lceil \frac{L}{p} \right\rceil, \quad (14)$$

then after necessary boundary padding, the sequence can be represented as:

$$\mathbf{C}^{(p)} \in \mathbb{R}^{K_p \times p \times 1}. \quad (15)$$

Here, K_p represents the number of temporal blocks contained in the historical window, and p represents the number of phase positions within each block. Each block is flattened into a token:

$$\mathbf{q}_k^{(p)} = \text{Flatten}(\mathbf{C}_{k,:,:}^{(p)}) \in \mathbb{R}^p. \quad (16)$$

For multiple screened candidate scales collected in $\mathcal{P}$, the model organizes short-term temporal patterns at different local phase resolutions.

Router-based Phase Attention.

Each phase token first undergoes linear projection to obtain a d_m -dimensional representation, and learnable position encoding is added:

$$\mathbf{e}_k^{(p)} = \text{LN} \left(\mathbf{W}_{\text{in}}^{(p)} \mathbf{q}_k^{(p)} \right) + \mathbf{p}_k^{(p)}. \quad (17)$$

To aggregate context across different temporal blocks, the model introduces R learnable router tokens. Let:

$$\mathbf{R}^{(p)} \in \mathbb{R}^{R \times d_m}, \quad (18)$$

then the routers first extract global information from the temporal-scale tokens:

$$\mathbf{C}^{(p)} = \text{Attn} \left(\mathbf{R}^{(p)}, \mathbf{E}^{(p)}, \mathbf{E}^{(p)} \right), \quad (19)$$

Subsequently, the temporal-scale tokens read cross-block context from the router representations:

$$\tilde{\mathbf{E}}^{(p)} = \text{Attn} \left(\mathbf{E}^{(p)}, \mathbf{C}^{(p)}, \mathbf{C}^{(p)} \right). \quad (20)$$

With residual connections, layer normalization, and feed-forward networks, router tokens summarize cross-block context without full pairwise interaction among all temporal-scale tokens.

Multi-scale common wind speed prediction.

For each candidate temporal scale, the model concatenates the last block token with the average representation of all block tokens:

$$\mathbf{s}^{(p)} = \left[\tilde{\mathbf{e}}_{K_p}^{(p)}, \frac{1}{K_p} \sum_{k=1}^{K_p} \tilde{\mathbf{e}}_k^{(p)} \right]. \quad (21)$$

Subsequently, the future common wind speed is obtained through the temporal-scale branch prediction head:

$$\hat{\mathbf{c}}^{(p)} = f_{\text{head}}^{(p)}(\mathbf{s}^{(p)}) \in \mathbb{R}^H. \quad (22)$$

The prediction results of different candidate temporal scales are combined by learnable fusion weights:

$$\hat{\mathbf{c}} = \sum_{p \in \mathcal{P}} \beta_p \hat{\mathbf{c}}^{(p)}, \quad \beta_p = \frac{\exp(\gamma_p)}{\sum_{q \in \mathcal{P}} \exp(\gamma_q)}. \quad (23)$$

Thus, Phase-Attention learns a fusion of multiple short-term candidate scales rather than relying on a single fixed temporal organization.

4.4. Shared Wind Speed Residual Prediction and Reconstruction

Common wind speed prediction describes shared farm-level variation, while turbine-level residuals recover local dynamics. For the i -th turbine, the residual historical sequence is:

$$\mathbf{r}_i = [r_{1,i}, r_{2,i}, \dots, r_{L,i}]. \quad (24)$$

The residual branch uses a shared MLP to predict local deviations. Taking the last observed residual $r_{L,i}$ as a reference, the centered sequence is:

$$\tilde{r}_{t,i} = r_{t,i} - r_{L,i}. \quad (25)$$

The shared residual predictor outputs the future residual variation:

$$\Delta \hat{\mathbf{r}}_i = f_{\text{MLP}}(\tilde{\mathbf{r}}_i). \quad (26)$$

The final residual prediction is:

$$\hat{r}_{h,i} = r_{L,i} + \Delta \hat{r}_{h,i}, \quad h = 1, \dots, H. \quad (27)$$

In the standardized wind speed space, the model uses turbine-level learnable scaling s_i and bias b_i to reconstruct the future wind speed:

$$\hat{\hat{v}}_{h,i} = s_i \hat{c}_h + b_i + \hat{r}_{h,i}. \quad (28)$$

Finally, $\hat{\hat{v}}_{h,i}$ is unstandardized to obtain the wind speed prediction in the physical space.

4.5. Wind Direction Independent Patch Transformer

Two-dimensional direction input.

For the i -th turbine, the historical wind direction sequence is represented as:

$$\mathbf{U}_i = [\mathbf{u}_{1,i}, \mathbf{u}_{2,i}, \dots, \mathbf{u}_{L,i}], \quad \mathbf{u}_{t,i} = [\cos \theta_{t,i}, \sin \theta_{t,i}]. \quad (29)$$

Accordingly, the direction branch does not extract a common wind direction or construct an additive “common direction + direction residual” decomposition. Each turbine direction sequence is maintained as an independent two-dimensional direction input.

Temporal patch partitioning.

For the $L \times 2$ direction sequence of each turbine, a sliding window with length P and step size S constructs temporal patches. The m -th patch is:

$$\mathbf{z}_{m,i} = \text{Flatten} \left(\mathbf{U}_{i,(m-1)S+1:(m-1)S+P} \right) \in \mathbb{R}^{2P}. \quad (30)$$

The number of generated patches is:

$$M = \left\lfloor \frac{L - P}{S} \right\rfloor + 1. \quad (31)$$

Each patch contains the sine and cosine components of P time points and is projected to a d -dimensional token with positional encoding:

$$\mathbf{e}_{m,i} = \mathbf{W}_{\text{patch}} \mathbf{z}_{m,i} + \mathbf{p}_m. \quad (32)$$

This patch representation captures local turning patterns and longer-range directional evolution [25, 31].

Shared Transformer encoding and prediction.

During direction encoding, each turbine is treated as an independent two-dimensional direction sequence. Turbines share the same patch projection, positional encoding, and Transformer encoder, without token mixing across turbines.

The encoded representation is aggregated using the last patch token and the mean token:

$$\mathbf{h}_i = \left[\mathbf{e}_{M,i}^{\text{enc}}; \frac{1}{M} \sum_{m=1}^M \mathbf{e}_{m,i}^{\text{enc}} \right]. \quad (33)$$

The prediction head maps $\mathbf{h}_i$ to the two-dimensional direction components for the future H time steps:

$$\hat{\mathbf{U}}_i = f_{\theta}(\mathbf{h}_i) \in \mathbb{R}^{H \times 2}. \quad (34)$$

The predicted vectors are L2-normalized before angle recovery:

$$\hat{\mathbf{u}}_{h,i} = \frac{\hat{\mathbf{u}}_{h,i}}{\|\hat{\mathbf{u}}_{h,i}\|_2 + \epsilon}. \quad (35)$$

Then atan2 recovers the wind direction angle.

4.6. Joint Loss Function

Wind speed loss.

Wind speed prediction uses Mean Squared Error in the standardized space:

$$\mathcal{L}_v = \text{MSE}(\hat{\mathbf{V}}, \tilde{\mathbf{V}}). \quad (36)$$

Direction-vector loss.

For the predicted direction and true direction, unit vectors are obtained respectively:

$$\hat{\mathbf{u}}_{t,i} = [\hat{u}_{t,i}^{\cos}, \hat{u}_{t,i}^{\sin}], \quad \mathbf{u}_{t,i} = [u_{t,i}^{\cos}, u_{t,i}^{\sin}]. \quad (37)$$

The direction error is defined using two-dimensional vector consistency:

$$\ell_{\theta,t,i} = 1 - \hat{\mathbf{u}}_{t,i}^{\top} \mathbf{u}_{t,i}. \quad (38)$$

At low wind speeds, wind direction observations are more susceptible to noise; therefore, a reliability weight based on physical wind speed is used:

$$w_{t,i} = \text{clip}\left(\frac{v_{t,i}}{v_0}, 0, 1\right), \quad (39)$$

where v_0 is the wind-speed threshold controlling the reliability weight. The weighted direction loss is:

$$\mathcal{L}_\theta = \frac{\sum_{t,i} w_{t,i} \ell_{\theta,t,i}}{\sum_{t,i} w_{t,i} + \epsilon}. \quad (40)$$

This loss constrains the angle between predicted and true two-dimensional direction vectors and avoids boundary errors from scalar-angle regression.

Wind vector consistency loss.

To constrain speed-direction consistency, physical wind speed is combined with the unit direction vector:

$$\mathbf{q}_{t,i} = v_{t,i} [\cos \theta_{t,i}, \sin \theta_{t,i}], \quad (41)$$

After scale normalization, predicted and true wind vectors are compared by Smooth L1 loss:

$$\mathcal{L}_q = \text{SmoothL1} \left(\frac{\hat{\mathbf{q}}}{\bar{\sigma}_v}, \frac{\mathbf{q}}{\bar{\sigma}_v} \right), \quad (42)$$

where $\bar{\sigma}_v$ is the average of the wind speed standard deviations of all turbines.

Overall objective.

The final training objective is:

$$\mathcal{L} = \lambda_v \mathcal{L}_v + \lambda_\theta \mathcal{L}_\theta + \lambda_q \mathcal{L}_q. \quad (43)$$

The coefficients balance the wind-speed objective, the direction-vector objective, and the auxiliary wind-vector consistency constraint.

5. Experiments

5.1. Experimental Setup

We evaluate WindFormer on two SCADA-based wind farm datasets with distinct spatial layouts and wind regimes: Shanxi with 24 turbines and Penmanshiel with 14 turbines. Both datasets are sampled at 10-min intervals and preprocessed following the protocol in Section 3. Unless otherwise specified, the input window is 288 steps, corresponding to 48 h of history, and the forecasting horizons are 6, 12, and 18 steps.

3D Visualization of Wind Turbine Spatial Distribution

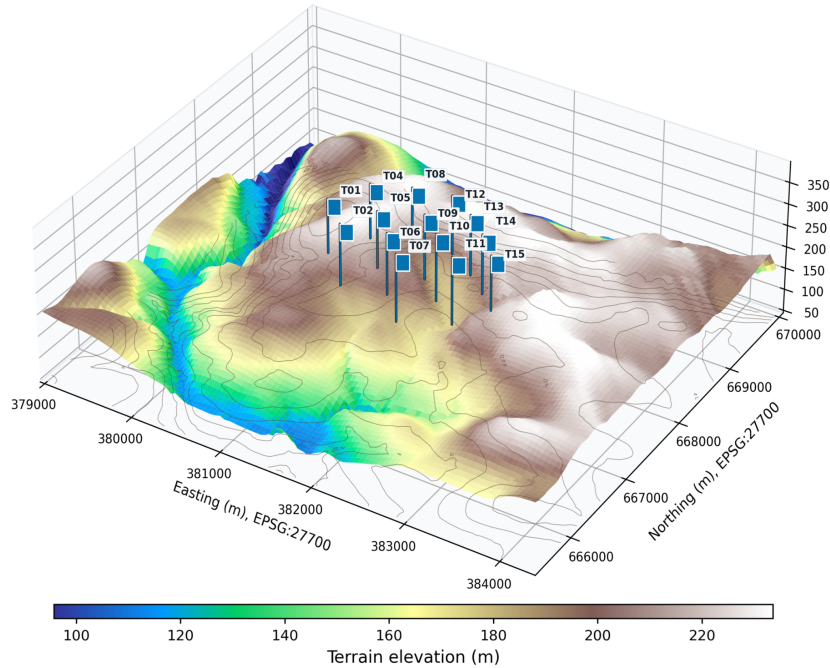


Figure 5: Geographical visualization of the two wind farm datasets used in the experiments.

WindFormer is compared with MST-Net, PatchTST, TimeMixer++,

GLPT, DLinear, TimesNet, and iTransformer. Mean absolute error (MAE) and root mean squared error (RMSE) are reported for wind speed (m/s) and wind direction (rad). The experiments are organized around four research questions: standard accuracy and efficiency (RQ1), component attribution (RQ2), robustness to perturbations and extreme volatility (RQ3), and transfer to unseen turbines (RQ4).

5.2. RQ1: Does WindFormer Improve Forecasting Accuracy and Efficiency?

The standard setting trains and tests on the full turbine set without artificial perturbation. We first compare the complete forecasting performance on the two wind farms, then examine direction-aware error profiles, computational cost, and matched-seed statistical consistency.

Table 1: Detailed full-sample forecasting performance on the Penmanshiel dataset under the noise-free setting. Lower values indicate better performance.

Metric	Horizon	WindFormer	MST-Net	TimeMixer++	iTransformer	PatchTST	TimesNet	GLPT	DLinear
Speed MAE	6	0.7142	0.7978	0.9229	0.9272	0.9189	0.9210	0.9318	0.9263
	12	0.8812	0.9879	1.1421	1.1491	1.1347	1.1406	1.1543	1.1473
	18	1.0103	1.1249	1.2994	1.3081	1.2891	1.2976	1.3138	1.3061
Speed RMSE	6	1.1497	1.2162	1.3053	1.3118	1.2987	1.3033	1.3174	1.3108
	12	1.4211	1.5538	1.6162	1.6247	1.6054	1.6142	1.6325	1.6224
	18	1.6216	1.7599	1.8367	1.8477	1.8263	1.8348	1.8553	1.8447
Direction MAE	6	0.1456	0.1637	0.1898	0.1911	0.1884	0.1894	0.1926	0.1939
	12	0.1881	0.2113	0.2439	0.2460	0.2403	0.2434	0.2483	0.2497
	18	0.2204	0.2471	0.2847	0.2872	0.2807	0.2840	0.2899	0.2913
Direction RMSE	6	0.2914	0.3092	0.3314	0.3341	0.3294	0.3320	0.3354	0.3375
	12	0.3686	0.3905	0.4176	0.4213	0.4146	0.4184	0.4237	0.4260
	18	0.4244	0.4599	0.4796	0.4843	0.4761	0.4807	0.4870	0.4906

Tables 1 and 2 show that WindFormer achieves the best wind-speed and wind-direction accuracy across all reported horizons on both datasets. The consistent ranking across terrain conditions and forecast lengths supports the

Table 2: Detailed full-sample forecasting performance on the Shanxi dataset under the noise-free setting. Lower values indicate better performance.

Metric	Horizon	WindFormer	MST-Net	TimeMixer++	iTransformer	PatchTST	TimesNet	GLPT	DLinear
Speed MAE	6	0.6367	0.7062	0.8192	0.8222	0.8165	0.8274	0.8270	0.8217
	12	0.8025	0.8939	1.0343	1.0395	1.0309	1.0452	1.0447	1.0379
	18	0.9309	1.0337	1.1970	1.2024	1.1953	1.2098	1.2094	1.2013
Speed RMSE	6	0.9884	1.0377	1.1203	1.1248	1.1172	1.1320	1.1311	1.1242
	12	1.2374	1.3033	1.4035	1.4097	1.3986	1.4178	1.4167	1.4078
	18	1.4116	1.4862	1.6002	1.6073	1.5987	1.6172	1.6161	1.6059
Direction MAE	6	0.2717	0.2984	0.3460	0.3478	0.3445	0.3507	0.3500	0.3516
	12	0.3441	0.3806	0.4403	0.4426	0.4384	0.4465	0.4459	0.4476
	18	0.4019	0.4437	0.5130	0.5160	0.5115	0.5206	0.5199	0.5223
Direction RMSE	6	0.5530	0.5778	0.6225	0.6269	0.6210	0.6283	0.6273	0.6300
	12	0.6525	0.7041	0.7356	0.7409	0.7342	0.7454	0.7440	0.7471
	18	0.7275	0.7831	0.8196	0.8255	0.8184	0.8295	0.8285	0.8324

central design choice of matching the representation to the physical structure of each variable.

Figure 6 shows that the advantage is not restricted to a single aggregate score: the error profiles remain favorable across wind directions, wind variables, datasets, and reported metrics.

The representative forecasts in Fig. 7 further illustrate that WindFormer follows the shared trajectory while retaining turbine-level variation in the predicted signals.

Table 3: Computational efficiency comparison at the 12-step forecasting horizon.

Metric	WindFormer	MSGNet	PatchTST	iTransformer	TimeMixer++
Parameters	36,512	26,680,000	140,492	124,428	388,332
Train time (s)	143.78	14356.30	315.32	57.80	109.55
Peak GPU (MB)	74.83	4532.1	3158.9	109.1	308.9

Table 3 shows that WindFormer combines the strongest forecasting re-

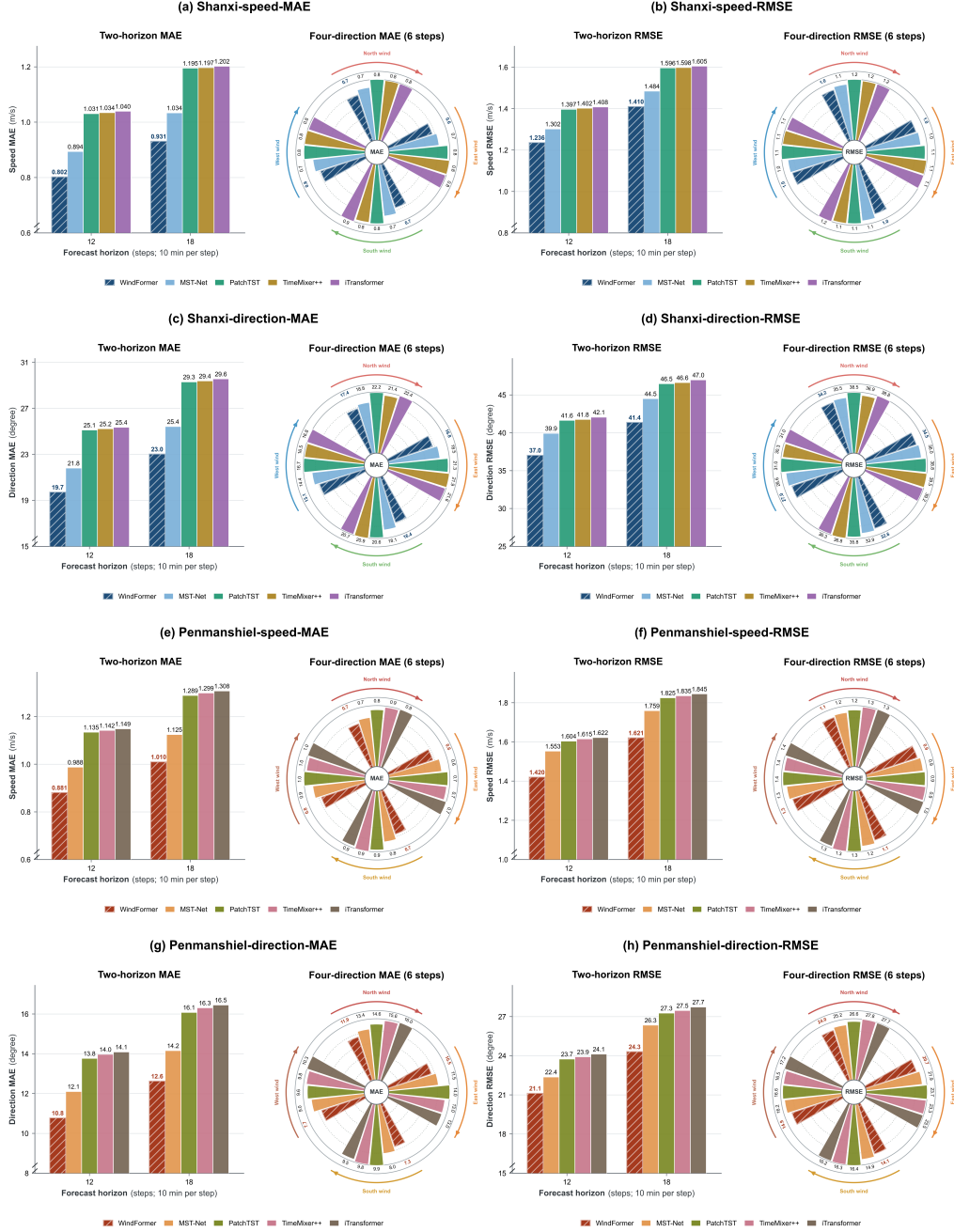


Figure 6: Direction-aware forecasting results across the Shanxi and Penmanshiel wind farms. Each panel reports the two-horizon error comparison and four-direction error profile for one dataset–metric pair.

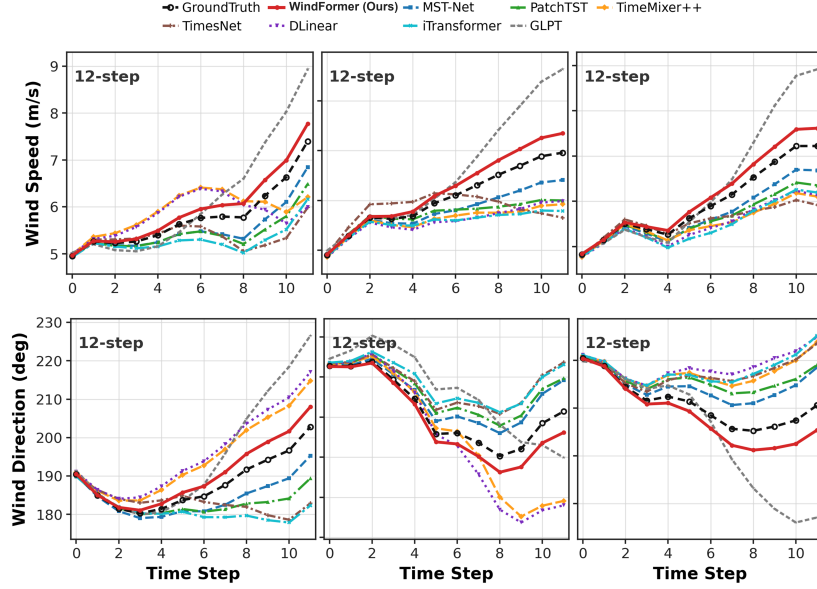


Figure 7: Representative 12-step full-sample prediction examples for wind speed and wind direction.

sults with a compact 36,512-parameter model and low peak GPU memory in the reported comparison.

The paired tests in Table 4 compare WindFormer against MST-Net over five matched random seeds. All eight dataset–metric comparisons remain significant after Holm correction, supporting the consistency of the observed improvements rather than attributing them to a single seed.

Together, these results answer RQ1 affirmatively. WindFormer improves accuracy across both wind variables and datasets while remaining computationally compact and statistically consistent across matched seeds.

Table 4: Statistical significance of WindFormer improvements over MST-Net across five matched random seeds. Values are mean $\pm$ sample SD. The one-sided paired t -test evaluates whether WindFormer has lower error; p_{raw} and p_{Holm} are the raw and Holm-adjusted values across the eight dataset-metric comparisons.

Dataset	Metric	WindFormer	MST-Net	p_{raw}	p_{Holm}
Shanxi	Speed MAE	0.8025 ± 0.0048	0.8939 ± 0.0072	2.94×10^{-5}	1.76×10^{-4}
	Speed RMSE	1.2374 ± 0.0074	1.3033 ± 0.0105	4.22×10^{-4}	6.60×10^{-4}
	Direction MAE	0.3441 ± 0.0027	0.3806 ± 0.0038	3.56×10^{-5}	1.76×10^{-4}
	Direction RMSE	0.6525 ± 0.0052	0.7041 ± 0.0070	3.30×10^{-4}	6.60×10^{-4}
Penmanshiel	Speed MAE	0.8812 ± 0.0053	0.9879 ± 0.0078	3.02×10^{-6}	2.41×10^{-5}
	Speed RMSE	1.4211 ± 0.0085	1.5538 ± 0.0124	1.73×10^{-5}	1.21×10^{-4}
	Direction MAE	0.1881 ± 0.0015	0.2113 ± 0.0021	3.48×10^{-5}	1.76×10^{-4}
	Direction RMSE	0.3686 ± 0.0030	0.3905 ± 0.0039	1.46×10^{-4}	4.37×10^{-4}

5.3. RQ2: Which Structural Components Drive the Improvements?

RQ2 examines whether the gains are attributable to the components introduced for the two challenges rather than to model capacity alone. The ablations therefore remove the learnable common-field weights, the circular direction representation, the wind-vector consistency loss, or the Phase-Attention temporal organization.

Table 5: Ablation study results for different variants on the Shanxi and Penmanshiel datasets.

Variant	Shanxi-6steps				Shanxi-12steps				Penmanshiel-6steps				Penmanshiel-12steps			
	Speed		Wdir		Speed		Wdir		Speed		Wdir		Speed		Wdir	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
WindFormer	0.6367	0.9884	0.2717	0.5530	0.8025	1.2374	0.3441	0.6525	0.7142	1.1497	0.1456	0.2914	0.8812	1.4211	0.1881	0.3686
w/o Learnable Weights	0.6754	1.0421	0.2765	0.5601	0.8465	1.3012	0.3495	0.6612	0.7589	1.2185	0.1502	0.2985	0.9325	1.5052	0.1935	0.3768
w/o Direction Vector Rep.	0.6502	1.0065	0.6214	1.0851	0.8188	1.2615	0.7245	1.2562	0.7285	1.1712	0.4985	0.8124	0.8985	1.4485	0.5642	0.9351
w/o Wind Vector Cons.	0.6612	1.0215	0.2985	0.5921	0.8315	1.2825	0.3852	0.7124	0.7412	1.1921	0.1685	0.3341	0.9125	1.4721	0.2185	0.4215
Phase-Attn to Patch	0.6845	1.0563	0.2798	0.5644	0.8582	1.3155	0.3541	0.6658	0.7695	1.2312	0.1534	0.3018	0.9458	1.5215	0.1972	0.3812

Table 5 supports the intended division of labor. Removing the direction-

vector representation mainly harms direction prediction, whereas removing the learnable common-field weights or replacing Phase-Attention with ordinary patches affects wind-speed prediction more strongly. Removing the wind-vector consistency term degrades both tasks, consistent with its role as the objective-level coupling between the two branches.

5.4. Hyperparameter Sensitivity Analysis

We further evaluate the sensitivity of WindFormer to four capacity and receptive-field choices: the historical input window length, the Phase Token projection dimension, the hidden dimension of the shared residual MLP, and the number of direction-branch Transformer layers. The results are summarized in Fig. 8. Across all four metrics, the default configuration remains close to the best-performing setting, indicating that the structural design is not dependent on a narrow hyperparameter choice.

These results answer RQ2 by linking each component to the challenge it was designed to address, while showing that the complete model does not depend on a single finely tuned capacity setting.

5.5. RQ3: Is WindFormer Robust under Noisy and Extreme Conditions?

RQ3 evaluates whether the structure-aware design remains useful when the inputs deviate from the clean training distribution. We consider sensor perturbations and increasing extreme-volatility severity.

For sensor perturbation, let $\theta_{i,t}$ denote the wind direction of turbine i at time t , and let $\mathcal{W}(\cdot)$ wrap angles to the valid range. Only the wind-direction channel in the training inputs is corrupted as

$$\tilde{\theta}_{i,t}^{(\sigma)} = \mathcal{W}\left(\theta_{i,t} + \epsilon_{i,t}^{(\sigma)}\right), \quad \epsilon_{i,t}^{(\sigma)} \sim \mathcal{N}(0, \sigma^2), \quad (44)$$

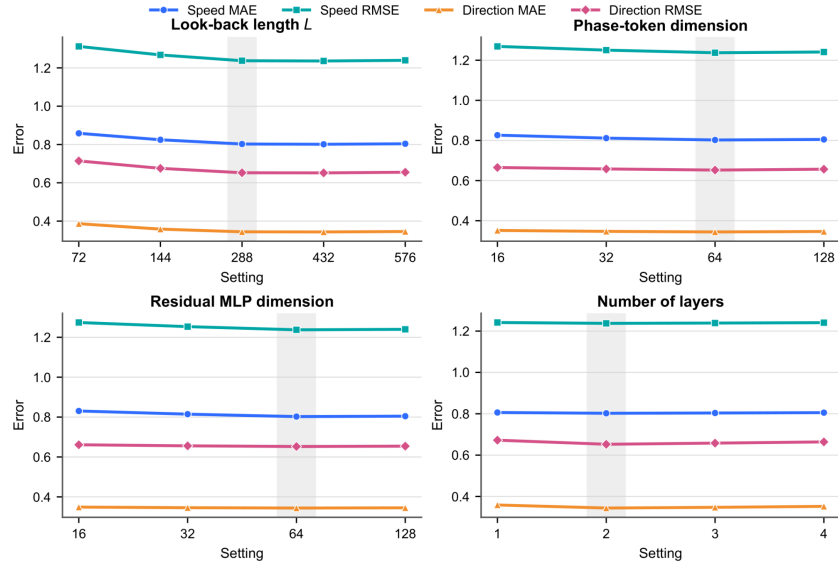


Figure 8: Hyperparameter sensitivity analysis on the Shanxi dataset. The four panels vary the historical window length, Phase Token projection dimension, shared residual MLP width, and direction-branch Transformer depth.

where $\sigma \in \{0.1, 0.2, 0.3\}$ rad. Test inputs and labels remain unchanged, and all models are retrained at each noise level.

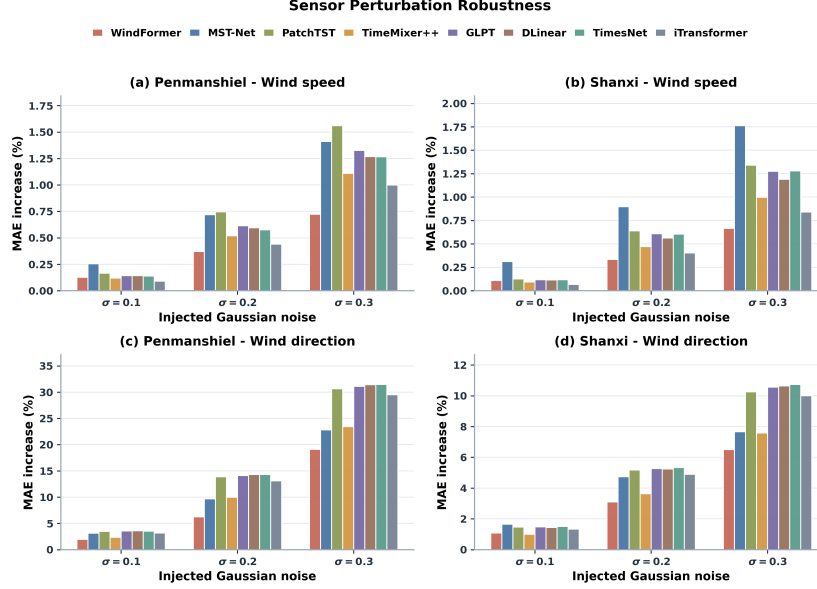


Figure 9: Sensor Perturbation Robustness. The comparison covers all models, both wind speed and wind direction, and both datasets. Each subplot uses its own vertical scale to preserve dataset-specific variation.

The second experiment evaluates gust-like transient fluctuations through an EVT-inspired residual stratification scheme [44, 45]. For a target series y_t , seasonal-trend decomposition using LOESS gives

$$y_t = T_t + S_t + R_t, \quad (45)$$

where T_t , S_t , and R_t denote the trend, seasonal, and residual terms, respectively. The extreme threshold is estimated only from the training split:

$$u = Q_{0.99}(\{|R_t| : t \in \mathcal{T}_{\text{train}}\}), \quad (46)$$

where $Q_{0.99}(\cdot)$ is the 99th percentile. A time step is identified as an extreme event when

$$I_t = \mathbb{I}(|R_t| > u). \quad (47)$$

Given an input window of length $L = 288$ ending at time t , the volatility severity of the corresponding forecasting sample is quantified by

$$N_{\text{extreme}}(t) = \sum_{\tau=t-L+1}^t I_{\tau}. \quad (48)$$

Test samples are binned by $N_{\text{extreme}}(t)$ and evaluated with the trained model held fixed.

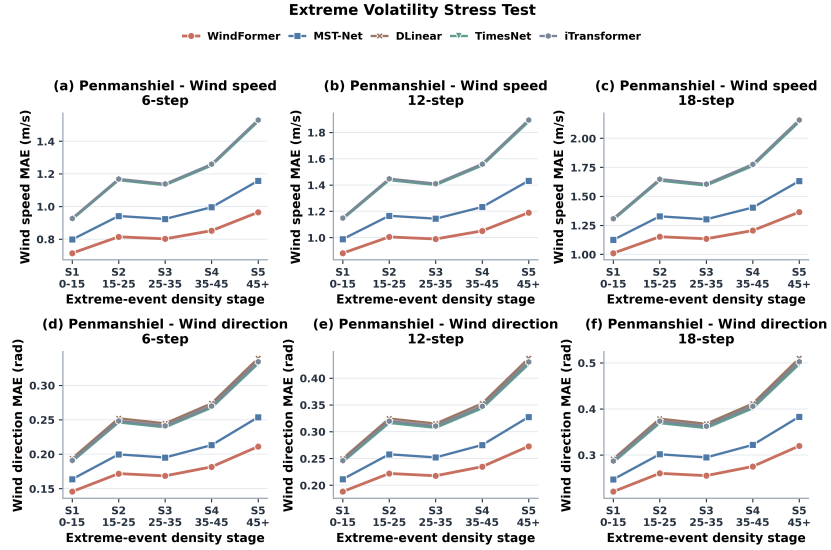


Figure 10: Extreme Volatility Stress Test on the Penmanshiel dataset. The figure reports wind speed and wind direction errors under increasing values of N_{extreme} .

Figures 9 and 10 show that WindFormer experiences the smallest degradation under both sensor noise and increasing extreme residual severity. This suggests that separating shared dynamics from local variation and respect-

ing directional geometry remains beneficial when the input distribution is perturbed.

These results answer RQ3 affirmatively: WindFormer retains a more stable error profile as measurement noise and transient volatility increase.

5.6. RQ4: Can WindFormer Generalize to Unseen Turbines?

RQ4 tests whether the shared temporal parameters can transfer beyond the turbines observed during training. We conduct the *Sparse Deployment Zero-Shot Evaluation* on Penmanshiel. Let $\mathcal{V}$ denote all turbines and $\mathcal{S}_k \subset \mathcal{V}$ the four turbines randomly selected in trial k . The remaining turbines

$$\mathcal{U}_k = \mathcal{V} \setminus \mathcal{S}_k \quad (49)$$

are treated as unseen zero-shot targets. For a model f_ϕ , ϕ is trained only on $\mathcal{D}_{\text{train}}(\mathcal{S}_k)$ and evaluated on $\mathcal{D}_{\text{test}}(\mathcal{U}_k)$. The reported error is averaged over three independent trials:

$$\text{Err}_{\text{ZS}} = \frac{1}{3} \sum_{k=1}^3 \text{Err}(f_{\phi_k}, \mathcal{D}_{\text{test}}(\mathcal{U}_k)). \quad (50)$$

Three conditions are compared: full-sample training, zero-shot transfer, and zero-shot transfer with 0.3-rad angular noise.

Figure 11 shows that WindFormer has the smallest zero-shot degradation, with a 17.72% increase in direction MAE under the clean setting and a 15.61% increase under 0.3-rad noise. The result indicates that shared temporal parameters and geometry-aware direction modeling can transfer to turbines that were not used as training targets.

Thus, RQ4 is answered affirmatively. WindFormer provides the strongest cross-turbine transfer among the compared models under the tested sparse-deployment conditions.



Figure 11: Sparse Deployment Zero-Shot Evaluation on Penmanshiel. The figure compares full-sample training, zero-shot transfer, and zero-shot transfer under sensor noise across multiple forecast horizons.

6. Conclusion

This paper presented WindFormer as a structure-aware framework for joint short-term forecasting of wind speed and wind direction in multi-turbine wind farms. Its central design principle is to share information where it is physically common and preserve information where it is turbine-specific: the wind-speed branch separates a learnable common field from local residuals and organizes the common history with Phase-Attention over screened candidate temporal scales, while the wind-direction branch retains turbine-wise sequences in a continuous sine-cosine space. A wind-vector consistency objective couples the two representation-decoupled branches during training.

The experiments answer the four research questions in sequence. WindFormer improves standard wind-speed and wind-direction forecasting across both public wind farms and remains computationally compact. The ablations show that the common-field weights, phase-aware temporal organization, circular direction representation, and vector-consistency loss contribute to the corresponding aspects of the performance. Perturbation and extreme-volatility tests show more stable degradation, while sparse-deployment experiments indicate transfer to unseen turbines. These findings support the broader conclusion that multi-turbine wind forecasting benefits from matching the model representation to the physical structure of each wind variable rather than applying one homogeneous interaction pattern to all channels.

The current study focuses on SCADA-based short-term forecasting and does not introduce a geographic graph correction in the final WindFormer configuration. Future work can investigate adaptive geographic or wake-aware extensions while retaining the same principle of separating shared dy-

namics from turbine-specific information.

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